

PROTOCOLS — ACPI PROTOCOLS

20.1 EFI_ACPI_TABLE_PROTOCOL

Summary

This protocol may be used to install or remove an ACPI table from a platform.

GUID

```
#define EFI_ACPI_TABLE_PROTOCOL_GUID \
    {0xffe06bdd, 0x6107, 0x46a6, \
     {0x7b, 0xb2, 0x5a, 0x9c, 0x7e, 0xc5, 0x27, 0x5c}}
```

Protocol Interface Structure

```
typedef struct _EFI_ACPI_TABLE_PROTOCOL {
    EFI_ACPI_TABLE_INSTALL_ACPI_TABLE    InstallAcpiTable;
    EFI_ACPI_TABLE_UNINSTALL_ACPI_TABLE  UninstallAcpiTable;
} EFI_ACPI_TABLE_PROTOCOL;
```

Parameters

InstallAcpiTable

Installs an ACPI table into the system.

UninstallAcpiTable

Removes a previously installed ACPI table from the system.

Description

The *EFI_ACPI_TABLE_PROTOCOL* provides the ability for a component to install and uninstall ACPI tables from a platform.

20.2 EFI_ACPI_TABLE_PROTOCOL.InstallAcpiTable()

Summary

Installs an ACPI table into the RSDT/XSDT.

Prototype

```

typedef
EFI_STATUS
(EFI_API *EFI_ACPI_TABLE_INSTALL_ACPI_TABLE) (
    IN EFI_ACPI_TABLE_PROTOCOL          *This,
    IN VOID                             *AcpiTableBuffer,
    IN UINTN                            AcpiTableBufferSize,
    OUT UINTN                           *TableKey,
);

```

Parameters

This

A pointer to a [EFI_ACPI_TABLE_PROTOCOL](#).

AcpiTableBuffer

A pointer to a buffer containing the ACPI table to be installed.

AcpiTableBufferSize

Specifies the size, in bytes, of the *AcpiTableBuffer* buffer.

TableKey

Returns a key to refer to the ACPI table.

Description

The *InstallAcpiTable()* function allows a caller to install an ACPI table. The ACPI table may either be a System Description Table or the FACS. For all tables except for the DSDT and FACS, a copy of the table will be linked by the RSDT/XSDT. For the FACS and DSDT, the pointer to a copy of the table will be updated in the FADT, if present.

To prevent namespace collision, ACPI tables may be created using UEFI ACPI table format, [UEFI ACPI Data Table](#). If this protocol is used to install a table with a signature already present in the system, the new table will not replace the existing table. It is a platform implementation decision to add a new table with a signature matching an existing table or disallow duplicate table signatures and return *EFI_ACCESS_DENIED*.

On successful output, *TableKey* is initialized with a unique key. Its value may be used in a subsequent call to *UninstallAcpiTable* to remove an ACPI table.

On successful output, the *EFI_ACPI_TABLE_PROTOCOL* will ensure that the checksum field is correct for both the RSDT/XSDT table and the copy of the table being installed that is linked by the RSDT/XSDT.

On successful completion, this function reinstalls the relevant *EFI_CONFIGURATION_TABLE* pointer to the RSDT.

Status Codes Returned

EFI_SUCCESS	The table was successfully inserted
EFI_INVALID_PARAMETER	The <i>AcpiTableBuffer</i> is <i>NULL</i> , the <i>TableKey</i> is <i>NULL</i> ; the <i>AcpiTableBufferSize</i> , and the size field embedded in the ACPI table pointed to by <i>AcpiTableBuffer</i> are not in sync.
EFI_OUT_OF_RESOURCES	Insufficient resources exist to complete the request.
EFI_ACCESS_DENIED	The table signature matches a table already present in the system and platform policy does not allow duplicate tables of this type.

20.3 EFI_ACPI_TABLE_PROTOCOL.UninstallAcpiTable()

Summary

Removes an ACPI table from the RSDT/XSDT.

Prototype

```
typedef
EFI_STATUS
(EFIAPI *EFI_ACPI_TABLE_UNINSTALL_ACPI_TABLE) (
    IN EFI_ACPI_TABLE_PROTOCOL    *This,
    IN UINTN                      TableKey,
);
```

Parameters

This

A pointer to a *EFI_ACPI_TABLE_PROTOCOL*.

TableKey

Specifies the table to uninstall. The key was returned from *InstallAcpiTable()*.

Description

The *UninstallAcpiTable()* function allows a caller to remove an ACPI table. The routine will remove its reference from the RSDT/XSDT. A table is referenced by the TableKey parameter returned from a prior call to *InstallAcpiTable()*.

On successful completion, this function reinstalls the relevant *EFI_CONFIGURATION_TABLE* pointer to the RSDT.

Status Codes Returned

EFI_SUCCESS	The table was successfully inserted
EFI_NOT_FOUND	TableKey does not refer to a valid key for a table entry.
EFI_OUT_OF_RESOURCES	Insufficient resources exist to complete the request.